3D Lineage-First Mycelial Fabric — Continual Discovery

Abstract

A persistent agent collective cannot synchronize every local memory into one canonical store, but merely preserving distributed memory is not enough either. We extend a lineage-first knowledge fabric — local evidence graphs, bounded shared abstractions, and typed derivation threads — with an endogenous epistemic loop in which agents generate strategic, domain-specific questions from the interaction of incomplete local perspectives, route those questions toward independently situated evidence holders, and use the resulting discoveries to determine what should be asked next. The central object is the *QuestionArtifact*: a bounded epistemic request that may travel even when raw private memory may not. The design principle we retain is *capability coherence without state convergence*. Our evidence spans three classes, which we keep separate throughout. Measured on a long-horizon conversational memory benchmark, routing a query to the correct evidence owner recovers oracle-level recall, and combined owner-routing with dialogue-pair back-fill improves end-to-end accuracy by a mean of 8.6 points under four independent graders; in the same setting, two independent interventions raise evidence availability while *lowering* answer accuracy, which is the empirical case for selective rather than exhaustive acquisition. In a deterministic repair fixture, lineage-targeted repair preserved more capability than replication at roughly half the transfer, though the fixture is too small to support a percentage-point claim. In a synthetic simulation of correlated evidence lineages, a combined questioning policy reached `0.871` root-balanced accuracy against 0.834 for the strongest baseline while spending about a third fewer questions, and the advantage of lineage-aware selection rose monotonically with lineage concentration — while being *worse* than random questioning at avoiding confident error in 15 of 23 cells. We report the negative results as part of the finding.

1 Introduction

Long-lived agent collectives create a systems problem that is easy to misname. A transcript is not a memory; a memory is not a knowledge claim; and a durable message stream is not a knowledge base. When these are collapsed, the resulting architecture is hard to reason about, hard to secure, and nearly impossible to evaluate causally.

The prior version of this work separated that stack. Each agent retains evidence-bearing memory in an independently owned local graph; policy-approved artifacts move through durable transport; and organizational knowledge is coordinated through a fabric with three typed relation families — horizontal peer coordination, vertical semantic abstraction, and lineage threads that preserve exact derivation paths. The design principle was *capability coherence without state convergence*: organizational knowledge can become more abstract at higher levels while remaining reconstructable from locally owned evidence.

That architecture is reactive. A query arrives, the router selects a coalition, evidence returns, an answer is synthesized. But a collective that only answers what it is asked cannot notice what it fails to know. It cannot detect that twenty agreeing agents descend from two upstream roots, one of which is corrupt. It cannot observe that a relationship exists between failures that no single agent is positioned to see. And it cannot decide which evidence would *discriminate* between competing explanations rather than merely corroborate the one it already holds.

This paper extends the fabric into an endogenous loop. Observations interact with local memory, temporal context, lineage, neighbouring abstractions, and contradictions to produce strategically valuable questions. Those questions route toward agents likely to hold informative and preferably independent evidence. Responses return with provenance. Discoveries, conflicts, and revisions generate the next question. The transition is from collective memory, to collective inquiry, to continual discovery.

Lineage is what makes this scientifically interesting rather than a matter of agents conversing. A conclusion may appear strongly supported because many agents repeat it; if all of them descend from one upstream source, the apparent consensus is not independent. Replica count is not independent epistemic support. A network that can detect this can ask for evidence specifically *outside* the dominant lineage — which is a different and more useful operation than retrieving more.

Contributions. (i) *QuestionArtifact*, a first-class bounded epistemic request that can traverse the fabric while raw private memory remains local. (ii) A continual discovery loop with five epistemic functions — verification, contradiction, relation discovery, hypothesis generation, and prediction/falsification — of which the earlier active-lineage-diversification controller becomes one special case. (iii) A sparsity mechanism: a value-driven priority function and budget regime intended to produce sparse strategic inquiry rather than constant chatter. (iv) An empirical account organized by *evidence class*, in which measured, simulated, and proposed results are never mixed, and in which the negative results carry as much weight as the positive ones.

2 System Model

Let agent i own a local memory graph $G_i = (V_i, E_i, \psi_i)$, where V_i is a set of persistent memory objects, E_i contains typed local relations, and ψ_i holds local metadata and policy state. There are no raw cross-graph edges; cross-node exchange occurs through typed artifacts.

A local memory $m \in V_i$ is an evidence-bearing state object. A knowledge claim k is a proposition whose truth for a task is supported by one or more authorized derivations over memories, $D(k) = \{d_1, \dots, d_r\}$ with $d_j : M_j \Rightarrow k$. Knowledge need not be stored as a literal sentence: a collective may know k if it can reconstruct k from distributed evidence when asked. A materialized KnowledgeArtifact $a = (\mathrm{id}, \mathrm{owner}, z, \Sigma, \tau, \lambda, p, \mathrm{refs})$ is an optimization and policy boundary, where z is semantic state, Σ is scope, τ is temporal validity, λ is lineage, p is authorization metadata, and refs are opaque pointers back to local evidence.

2.1 QuestionArtifacts

We introduce a second first-class object. A QuestionArtifact is

$$Q = (\mathrm{id},\ \mathrm{asker},\ q,\ \mathrm{trigger},\ \Sigma,\ \tau,\ u,\ \lambda_{\mathrm{ctx}},\ \mathscr{D},\ \pi,\ B,\ \mathrm{status}), \tag{1}$$

where q is the question, trigger records what produced it, u is the asker's uncertainty, λ_{ctx} is the lineage context that motivated it, $\mathscr{D}$ is a set of candidate domains, π is policy, and B is a budget.

QuestionArtifacts are not memories. They are bounded epistemic requests, and the distinction has an architectural consequence: **questions may travel; raw private memory does not have to.** An agent can reveal that it is asking what evidence exists for X without exposing its local store. Responses return carrying provenance, so the asker learns what was observed, who contributed, which evidence roots were independent, when the evidence was created, what derivation produced the conclusion, and where disagreement remains.

3 The 3D Lineage Fabric

The fabric is a multilayer directed knowledge graph $M = (V, E_H, E_V, E_L, \ell, \Sigma, \tau, \pi)$, where $\ell(v)$ is the organizational abstraction level, $\Sigma(v)$ the set of overlapping scopes attached to v, $\tau(v)$ temporal validity, and $\pi(v)$ policy state. The horizontal plane E_H answers who at the same abstraction level can contribute; the vertical axis E_V determines what lower-level evidence deserves to become a higher-level representation; and lineage threads E_L record exactly which lower-level evidence and transformations support a higher-level artifact, forming a derivation DAG that makes abstraction

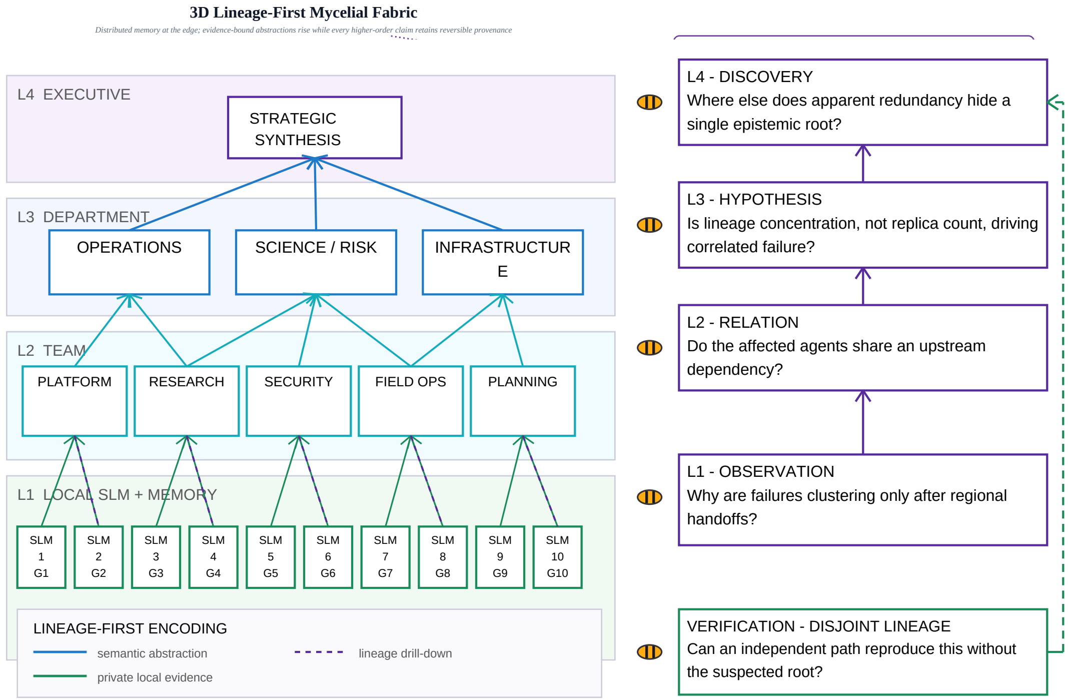

Figure 1: The 3D Lineage-First Mycelial Fabric carrying a strategic question chain. The fabric is unchanged from the baseline architecture: local agents hold private evidence graphs, abstraction rises vertically, and lineage threads keep every higher-order claim drillable to independent lower-level evidence. What is added is endogenous inquiry. Questions escalate in cognitive abstraction rather than merely in scope — from an observation, to a relation between agents, to a network hypothesis, to a strategic discovery — and the loop closes when a verification question is routed back to a region deliberately disjoint from the lineage under suspicion, so that agreement cannot be inherited from the source being tested.

reversible enough for audit, drill-down, and repair. Cross-functional scopes such as projects and missions remain overlapping hyperedges rather than being forced into the organization tree.

Figure 1 shows the fabric carrying a question chain. The progression matters as much as the mechanism: questions become broader, more causal, and more counterfactual as evidence is abstracted, and the loop closes when a verification question is routed back toward a region disjoint from the lineage under suspicion.

3.1 Independent support versus copied support

Let P_k be the set of valid authorized derivation paths that reconstruct k, each path p depending on a set of failure domains $D(p)$. A survivability margin is the minimum-cost set of failures intersecting every surviving derivation,

$$\kappa(k) \;=\; \min_{F \subseteq \mathscr{F}} \{\, c(F) : \forall p \in P_k,\; D(p) \cap F \neq \emptyset \,\}. \tag{2}$$

Equation 2 is deliberately recognizable as a hitting-set object; the novelty is not the formalism but what it is applied to. Ten memories descending from one upstream source have replica count ten and epistemic support one. For two derivations, strict independence with respect to a selected domain family $\mathscr{F}^*$ is $D_{\mathscr{F}^*}(p_i) \cap D_{\mathscr{F}^*}(p_j) = \emptyset$. Independence is domain-relative in practice: two paths may use different agents but the same database, or different documents written from the same human source. The fabric therefore records typed derivation edges rather than flattening provenance into an undifferentiated list of source identifiers.

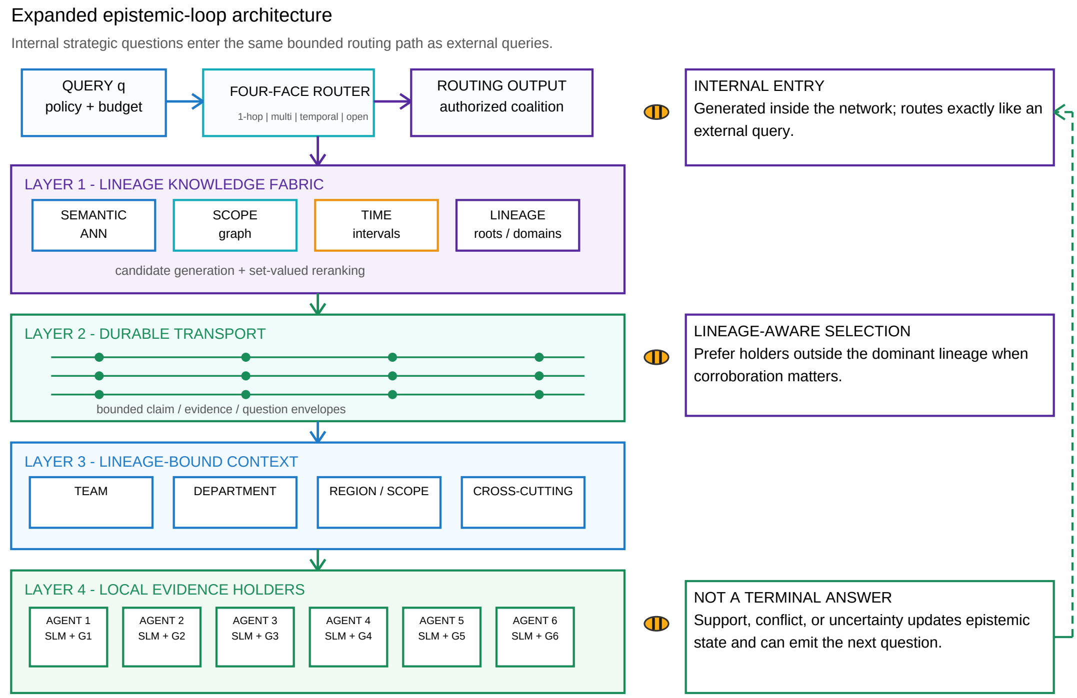

Figure 2: Expanded architecture with the epistemic loop closed. The baseline path — routing, lineage-aware knowledge fabric, durable transport, organizational context, local evidence holders — is preserved exactly. Two things change. An internally generated strategic question enters the same routing path as an external query, so questions become routable first-class objects rather than control flow. And synthesis no longer terminates: every outcome, including support, updates an epistemic state that can emit the next QuestionArtifact. Raw local memory never crosses the transport boundary; only bounded claim, evidence, and question envelopes do.

4 Continual Discovery

The reactive path terminates at an answer. The extended architecture terminates temporarily at an *updated epistemic state*, which can immediately generate another QuestionArtifact. Figure 2 shows the two entry points — an external query and an internally generated strategic question — entering compatible routing, and the feedback edge that closes the loop.

Continual discovery is not fact-checking. We distinguish five epistemic functions. **Verification** asks whether an independent path can reproduce a conclusion. **Contradiction** asks which observation cannot be explained by the current hypothesis. **Relation discovery** asks whether apparently unrelated failures share a hidden dependency. **Hypothesis generation** asks what mechanism would explain a set of distributed observations at once. **Prediction and falsification** ask which capability should fail next if the hypothesis holds, and which unaffected region should exhibit the pattern. The last is what makes the loop recursive rather than merely reactive: a prediction routed toward an unobserved region turns that region into a test.

Active lineage diversification as a special case. The earlier controller detected high-value claims with low $\kappa(k)$ and requested blind verification through a disjoint source, tool, or agent path, choosing $a^* = \arg\max_a \mathbb{E}[\kappa'(k) - \kappa(k) \mid a]/C(a)$. That controller is preserved and generalized: it is the verification branch of the loop above. When the system detects a high-value proposition with weak independent support, it emits a verification QuestionArtifact. The verifier must not receive the existing answer where doing so would contaminate independence; otherwise the mechanism produces correlated confirmation rather than independent support. Disagreement becomes an explicit conflict object rather than an automatic merge.

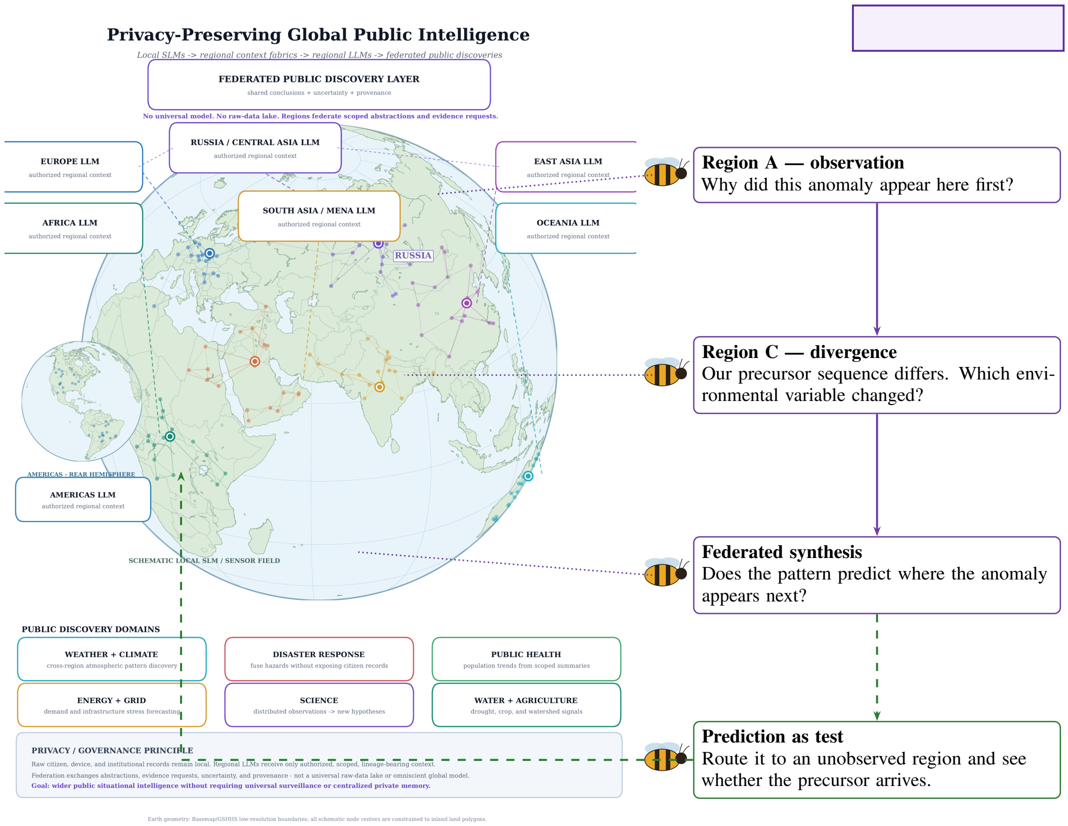

Figure 3: Federated public discovery under the same loop (future vision, schematic). Raw records stay local; regions federate scoped abstractions, uncertainty, and provenance. Continual discovery adds the final step: a synthesized prediction is routed to a region that has *not* yet observed the pattern, converting the prediction into a test. No result in this paper is measured on this configuration.

4.1 Deployment shape

Figure 3 shows the loop instantiated across independently governed regions. Raw citizen, device, and institutional records stay local; regions federate only scoped abstractions, uncertainty, and provenance. The step continual discovery adds is the last one: a synthesized prediction is routed toward a region that has *not* yet observed the pattern, which converts a hypothesis into a test rather than a broadcast. This is the configuration the architecture is aimed at. No result in this paper is measured on it, and we mark it as a schematic wherever it appears.

4.2 Question selection

A continually questioning network becomes computationally absurd if every agent asks everything. Questions must be sparse and value-driven. We propose a priority function

$$\text{Priority}(q) = \frac{\text{Value}(q) \cdot \text{Uncertainty}(q) \cdot \text{Impact}(q) \cdot \text{EpistemicGap}(q) \cdot \text{EIG}(q)}{\text{ExpectedCost}(q)}, \tag{3}$$

driven by signals including uncertainty, contradiction, lineage diversity, reconstruction margin, staleness, scope importance, expected information gain, novelty, prediction error, and the number of downstream claims affected. Explosion is controlled with budgets, rate limits, semantic duplicate suppression, question clustering, scope constraints, authorization, cooldowns, and priority thresholds. Equation 3 is *proposed*. It is not validated by any experiment in this paper; what the simulation in Section 5.4 does establish is that the *stopping* rule, not the selection rule, is where a lineage-aware policy currently underspends.

1,540
QUESTIONS

72.2%
744-Q E2E

+8.6pp
4/4 GRADERS

Class	Source	Scale	Standing
A. Measured	Conversational memory benchmark	1,540 questions	Real system, real runs
B. Repair fixture	Deterministic simulator	9 capabilities	Mechanism demonstration
C. Questioning	Synthetic simulation	10 seeds $\times$ 2,000 claims	Mechanism probe

Table 1: Evidence classes. Class A involves a real retrieval and answering system. Classes B and C are simulators: there are no agents, messages, or network in either. No claim in this paper is a deployment result.

Retriever	single-hop R@10	overall R@10	end-to-end acc.
Pure embedding	25.9	25.3	—
Typed retriever, pooled	39.4	34.9	64.9
+ speaker boost, embedding back-fill	44.7	37.2	67.0
+ graph neighbours appended	39.4	34.9	66.7
Pure embedding + owner routing	**46.8**	—	—
Typed + owner routing + pairs (shipped)	**46.8**	**46.8**	**73.8**

Table 2: Class A retrieval ladder. Owner-routing is the largest single step and matters most to the retriever with no structural representation of ownership.

5 Experiments

5.1 Evidence classes

The results below come from three sources with very different standing. We label them explicitly and never combine them in one table.

5.2 Class A: measured distributed retrieval

All Class A measurements use a long-horizon conversational memory benchmark: 10 conversations, 1,540 questions across single-hop (282), multi-hop (841), temporal (321), and open-domain (96) categories. Accuracy is an LLM-judge verdict against a gold answer; recall@k tests whether any part of the gold answer appears in the top k retrieved messages with no model calls. An earlier campaign used the gold answer as an acceptance gate and the gold category label for routing; both were removed, and the affected runs are excluded rather than retained as baselines.

We state the scope of the distributed claim plainly. Independent ownership is instantiated here in its minimal non-trivial form: each participant holds a separate store the other cannot read. This is two evidence holders, not a population.

End-to-end accuracy of the shipped configuration. On 744 benchmark questions the shipped stack — owner routing, pair back-fill, 80-candidate reranking, and 15 memories in the answer prompt — reaches 72.2% overall accuracy (single-hop 73.9, multi-hop 72.8, temporal 73.7, open-domain 56.5). On a 584-question subset scored by four graders the same configuration scores 73.8, 65.6, 42.5, and 26.7. We give the spread rather than a headline number, for the reason developed below.

The retrieval ladder. Table 2 shows where the gains actually come from. Owner-routing is the single largest step, and it is much larger for the semantic retriever (25.9 $\rightarrow$ 46.8) than for the typed one (39.4 $\rightarrow$ 46.8) — consistent with the identity probe below, since a pure embedding has no other way to represent who holds what. Budget-matched, 30 extra candidates drawn from graph neighbours add 1.4 recall@all while 30 drawn from plain embedding add 4.1, and 30 drawn from dialogue pairs add 3.2 over that control.

Routing to the correct owner matches oracle routing. Pooling all memories into one index is the natural baseline. Routing each query to the store of the named participant recovers almost exactly

+8.6pp

MEAN GAIN

53.7

R@10

78->68

MORE != BETTER

Retrieval condition	single-hop R@10	multi-hop R@10
Pooled index (control)	39.4	46.8
Routed to named holder	**46.8**	50.3
Oracle routing (gold owner)	46.5	51.0
Federated by per-store score	36.9	42.4

Table 3: Class A. Owner-routing recovers oracle-level recall. Score-level federation across independently owned stores underperforms simple pooling.

Memories in prompt	Accuracy	Gold present	Accuracy | gold present
5 / 10	69	57%	87.7
15 / 30 (shipped)	**78**	71%	**88.7**
30 / 30	76	71%	87.3
50 / 50	69	**73%**	78.1

Table 4: Class A, 100 matched single-hop questions. More context raises the chance the gold evidence is present and lowers accuracy by 9 points.

what an oracle with the gold owner label recovers (Table 3). The final row is the more interesting one: fusing candidates by raw per-store score is *worse* than pooling, because scores produced by independently maintained stores are not calibrated against one another. Independence of ownership is precisely what breaks naive score comparison, which is why the router selects a set of holders rather than merging their scores.

Selective acquisition improves answers. Combining owner-routing with dialogue-pair back-fill improves end-to-end accuracy on 282 single-hop questions by a margin that is positive under all four graders we tested: +8.9, +11.3, +9.6, and +4.6 points, mean +8.6. We attribute this to the combined selection-and-context mechanism rather than to routing alone — the treatment changes both which holder is consulted and which candidates are back-filled, and the two were not ablated apart end-to-end. This is the best-powered positive result in the paper.

The negative result: more evidence, worse answers. The natural next move — retrieve more — fails, through two independent mechanisms. Widening the answer context monotonically increases evidence availability while accuracy peaks and then falls (Table 4). The conditional column is the important one: accuracy *given that the gold evidence was present* falls by 10.6 points. The additional evidence did not merely fail to help; it degraded the system's use of evidence it already held.

Second, a stronger retriever reproduces the effect. Fusing vector search, an entity-graph hop, and pair expansion raises overall recall@10 from 46.8 to 53.7 across 1,219 questions. On the matched 100 single-hop identifiers, accuracy falls from 78 to 68 (3 wins against 13 losses); multi-hop is flat at 75 $\rightarrow$ 76. Accuracy given gold in context falls from 88.6 to 79.1. The measured cause is that the entity hop fills the candidate window with topically related non-answers that both the reranker and the answer model prefer. We note the asymmetry honestly: the recall gain is measured across three categories and the accuracy loss is concentrated in one.

Retrieving more of the right kind of thing is not the same as retrieving what discriminates. This is the empirical case for treating acquisition as a targeted epistemic act, and it is what motivates Section 4.

Provenance is not recoverable from semantics. A probe that rewrites each question with the other participant's name isolates how identity is encoded. Naming the *wrong* holder *improves* pure-embedding recall (26.6 $\rightarrow$ 34.8) and collapses identity-boosted retrieval (48.8 $\rightarrow$ 26.2). The measured reason is distributional: a participant's own name appears in 0.1% of their own messages and the other participant's in 33.4%. A name in a query is an embedding-level pointer toward the

18x

FAST PATH

7/9

REPAIR

5,448B

TRANSFER

wrong evidence holder. Provenance is not latent in the semantics of evidence and cannot be recovered by a better encoder; it must be carried as typed structure.

Measurement validity. Absolute accuracy on this benchmark is grader-dependent to a degree that makes cross-paper comparison unsafe. Agreement with the campaign judge is 86.5% ($\kappa = 0.72$) for a second lenient judge, 59.4% ($\kappa = 0.30$) under a strict rubric, and 49.1% ($\kappa = 0.17$) for substring matching. Judge model alone moves scores by roughly 12 points under an identical prompt; rubric leniency by roughly 30; the total spread across graders reaches 51 points. We therefore report every headline comparison under all four graders and treat only sign-stable effects as findings. Under that rule the routing result holds (4/4 graders), while an open-domain prompt intervention (+14.6 lenient, −1.0 strict) and an answer-model swap (sign flips with judge family) are recorded as null.

The reranker contributes ordering, not recall: gold-in-context is identical across every reranker variant, so retrieval decides what is reachable and reranking decides only what surfaces. Replacing pointwise with listwise scoring halves rerank latency for a 4-point accuracy loss, and removing the reranker entirely costs 6.5 points (68.6 → 61.9) while cutting end-to-end latency from 8.2 s to 0.44 s, an 18× speedup. Pair back-fill candidates occupy ranks 51–80 and never reach the prompt without reranking, which bounds how cheaply the fast path can currently be run.

5.3 Class B: the repair fixture

A deterministic simulator tests whether capability survives after evidence sources or shared derivation roots fail. Across a fixture of **nine** capabilities, lineage-targeted repair preserved 7, full replication and random diversification preserved 6 each, and source-count repair preserved 5, at serialized transfer volumes of 5,448 B, 9,877 B, 5,206 B, and 4,941 B respectively.

We report counts rather than percentages deliberately. With nine trials per condition the difference between lineage-targeted repair and full replication is a single capability, and a Fisher exact test on $7/9$ against $6/9$ returns $p = 1.00$; against $5/9$ it returns $p = 0.62$. The 95% Wilson interval on $7/9$ is $[45.3\%, 93.7\%]$ and overlaps every other condition. Detecting a difference of this size at 80% power would require roughly 127 capabilities per condition. **This fixture does not support a percentage-point performance claim, and we do not make one.**

What it does support is a mechanism demonstration. In the shared-root variant, where a common upstream root is removed, replica-based and source-count repair recover nothing while lineage-aware independent repair recovers fully. That outcome is close to true by construction — if every replica descends from the removed root, replica-based repair cannot recover — which is exactly why we present it as an illustration of the definition of independent support rather than as evidence of superiority over replication. The useful reading is directional: transfer volume was roughly half that of full replication at equal or better survival, which motivates the frontier claim without establishing it.

5.4 Class C: simulated continual questioning

A synthetic generative model produces propositions with a latent truth, a small number of upstream roots with Zipf-distributed mass, per-root corruption, and candidate agents inheriting one root each. Five questioning policies compete over a shared root-balanced aggregator, so a policy cannot win by inferring better — only by acquiring better evidence. The headline sweep is 10 seeds × 2,000 claims; a sensitivity sweep covers 23 cells at 8 seeds × 1,500 claims, seed-paired within cells.

Three findings are worth stating. First, the combined policy gains about four accuracy points over the strongest baseline while spending roughly a third fewer questions than exhaustive random questioning; at a budget of 2 it is more accurate than random questioning at a budget of 8, a 5.8× reduction in questions asked (paired $+0.0097 \pm 0.0032$, 7 of 8 seeds).

Second, uncertainty-triggered questioning — the policy most systems actually implement — buys accuracy but leaves false confident consensus exactly where asking nothing left it (0.028 against 0.028). Confidence-triggered questioning cannot fix confidently wrong, because it never fires there. That is the failure mode this line of work exists to attack.

5.5 All results, consolidated

Finding	Class	Result	Verdict
Shipped configuration, end to end	A	72.2% over 744 questions	headline
Owner-routing vs pooling	A	R@10 +7.4 typed, +20.9 embedding	works
Owner-routing + pair back-fill	A	+8.6 pp mean, 4/4 graders	works
Score-level federation	A	36.9 vs 39.4 pooled	hurts
Pair nodes as back-fill	A	+3.2 recall@all at equal budget	works
Speaker boost + embedding fill	A	+2.1 end to end	small
Graph-neighbour expansion	A	+1.4 recall vs +4.1 for embedding	no better
Listwise reranker	A	−4 acc, −56% rerank latency	speed only
Wider answer context	A	gold 71 → 73%, acc 78 → 69	hurts
Graph-expanded retrieval	A	R@10 46.8 → 53.7, acc 78 → 68	hurts
LLM-built entity graph	A	R@10 +5.0, acc 78 → 70	hurts
Identity in the query	A	wrong name *raises* recall 26.6 → 34.8	structural
Reranker removal	A	−6.5 pp for 18× speedup	frontier
Judge sensitivity	A	51-pt spread; κ 0.17–0.72	measurement
Open-domain flags / model swap	A	sign-unstable across graders	null
LLM vs regex query router	A	79 vs 78; 52% agreement	null
Scoring-formula fixes	A	within 1.5 pts either way	null
Reply-only pair mapping	A	−1 to −2 recall	hurts
Pair context in rerank / prompt	A	−7 / −3 / −12	hurts
Lineage-targeted repair	B	7/9 vs 6/9 vs 5/9 capabilities	underpowered
Transfer volume	B	5,448 B vs 9,877 B	directional
Shared-root removal	B	recovery 1.00 vs 0.00	by construction
Combined questioning policy	C	0.871 vs 0.834, ∼1/3 fewer questions	works
Question budget efficiency	C	budget 2 beats random at 8 (5.8×)	works
Lineage concentration axis	C	monotone −0.021 → +0.012, 0/8 → 8/8	works
Corruption-rate sweep	C	leads all 5 cells, +3.1 to +4.6 pts	works
Candidate-pool sweep	C	advantage widens 16 → 128	works
Independent-root target	C	lineage 0.802 → 0.862 at target 5	stopping rule
Uncertainty-only triggering	C	confident error unchanged (0.028)	fails
Combined policy calibration	C	worse than random in 15/23 cells	**negative**
Lineage selection alone	C	3 wins, 8 ties, 12 losses	**negative**
Conflict detection	C	best policy misses 1 claim in 5	**negative**

Table 7: Every completed experiment, by evidence class. Class A is measured on a real retrieval and answering system; B and C are simulators. Negative results are shown in the same table as positive ones deliberately.

Sensitivity and negative results.

The omitted per-sweep plots are summarized in Table 7. The combined questioning policy reaches 0.871 accuracy versus 0.834 for random while asking about one-third fewer questions; at budget 2 it exceeds random at budget 8. Lineage-aware selection becomes more useful as lineage concentration increases, but the combined policy is worse than random at avoiding confident error in 15/23 sensitivity cells. Conflict detection remains incomplete. These failures point to the stopping rule - not merely holder selection - as the main unresolved control problem.

6 Limitations, related work, and conclusion

The population-scale claim is not established: the measured retrieval study uses two evidence holders, the repair study is a nine-capability fixture, and the continual-questioning study is a synthetic correlated-evidence model rather than a deployed agent network. Experiments at 100, 1,000, and 10,000 agents remain future work. Prior work already covers graph memory, governed shared memory, lineage-aware enforcement, hierarchical organization, adaptive retrieval, and multi-agent routing; the contribution here is the combination and the measured phenomenon that lineage structure changes what evidence a collective should seek next. The central result is narrower than the architecture: where evidence came from matters more than how many copies exist, and selective questioning can improve accuracy and evidence efficiency - while poor stopping rules can increase confident error.

References

[1] P. Alvaro, J. Rosen, and J. Hellerstein. Lineage-driven fault injection. In *ACM SIGMOD*, 2015.

[2] A. N. Bala and F. M. Shah. Multi-agent routing as set-valued prediction: A WildChat benchmark and cost-aware evaluation. *arXiv:2606.28925*, 2026.

[3] H.-L. Hsu et al. Organize then retrieve: Hierarchical memory navigation for efficient agents. *arXiv:2606.11680*, 2026.

[4] S. Ji, Y. Li, and B. Hooi. Memory is reconstructed, not retrieved: Graph memory for LLM agents. *arXiv:2606.06036*, 2026.

[5] Y. Margalit et al. Governed shared memory for multi-agent LLM systems. *arXiv:2606.24535*, 2026.

[6] C. Ouyang and R. Hou. MemLineage: Lineage-guided enforcement for LLM agent memory. *arXiv:2605.14421*, 2026.

[7] Y. Wang et al. MAP-Graph: Provenance-aware shared memory for multi-agent workflows. *arXiv:2608.10509*, 2026.

[8] Y. Wang et al. OrgAgent: Organize your multi-agent system like a company. *arXiv:2604.01020*, 2026.

[9] Z. Wang et al. RAGRouter-Bench: A dataset and benchmark for adaptive RAG routing. *arXiv:2602.00296*, 2026.

[10] R. Wu et al. MemHarness: Memory is reconstructed, not replayed. *arXiv:2607.28272*, 2026.

[11] Y. Zheng et al. SkillRouter: Retrieve-and-rerank skill selection for LLM agents at scale. *arXiv:2603.22455*, 2026.

[12] X. Zhu et al. Can LLM agents sustain long-horizon organizational dynamics? *arXiv:2606.01199*, 2026.

9 Large-Scale Agentic Web Stress Test

Design. We stress-test lineage-first knowledge distribution on a simulated agentic web of N ∈ {10^2, 10^3, 10^4, 10^5} agents organised into teams, failure domains, organisations and regions. Knowledge is partial and local: every evidence item has a home agent, and each item descends from exactly one origin observation. Family sizes are heavy-tailed and ~35% of claims have a single origin, so replica count and independent evidential support come apart by construction. We evaluate 127,374 labelled questions per seed (single-hop, multi-hop, temporal, revision-sensitive, reconstruction, contradiction, evidence-verification, strategic ranking) against eight systems: isolated memory, a centralized store, full replication, random replication, gossip, a diversity-blind replica-count policy, the lineage-aware fabric, and the fabric with continual questioning. Equal-budget systems publish the identical number of replicas per claim (4); every probe attempt is charged as a message, including attempts on dead hosts. Failures escalate from 10–90% random churn to correlated loss of whole failure domains, organisations and regions, to a **white-box adversary** that knows each system's own placement and greedily removes the hosts of its scarcest surviving independent evidence, to partitions with independent updates, to corruption of up to 20% of nodes or of origin sources. 30 seeds, paired by seed, Holm-corrected.

Result. Under the white-box attack at N = 10,000 removing half the agents, knowledge survival is statistically indistinguishable across equal-budget policies (0.679 lineage, 0.678 random, 0.677 diversity-blind), but the lineage fabric is better on every provenance-sensitive quantity: task accuracy +0.010 (d_z = 2.29, Holm p <1e-6), independent-support survival +0.046 (d_z = 21.30), and contradiction F1 +0.155. The gap widens under origin-source corruption, where a corrupted origin's false value is inherited by all of its copies: accuracy +0.019 (Holm p <1e-6). Full replication survives better (0.870) at 194 replicas per claim — 48× the storage. The frontier (Fig. 2) therefore reads: at matched budget the three equal-budget policies coincide on survival and separate only on evidential quality, and the margin that broad replication holds over all of them is bought with two orders of magnitude more storage rather than with a better use of it.

Questioning ablation. At *identical storage* — the base budget is reduced by exactly the replicas questioning adds — continual questioning raises accuracy by +0.189 (d_z = 37.16, Holm p <1e-6) for a 1.81× increase in messages, and drives post-partition reconciliation (+0.314 accuracy after reconnection).

Negative findings. The same questioning machinery on a *count-based* system gains +0.245: continual discovery is largely orthogonal to lineage, not a consequence of it. Under corruption, re-verification can pull the false value from a corrupt origin (+0.165 accuracy). Lineage awareness does not improve raw survival under random churn (+0.000), cannot help the ~35% of claims with a single origin, and the placement × aggregation factorial shows that essentially all of the benefit comes from *which evidence is published*, not from lineage-weighted voting.

Table 1 (main results)

Table 1. Knowledge survival and downstream accuracy under four failure conditions at N = 10,000 (30 seeds, paired). KS is the fraction of valid knowledge that is both recoverable and answered correctly; ISS is the fraction of original independent evidence paths that survive in the system's own stored set; storage counts published replicas per claim and messages count every publication, question and probe attempt. B3, B5, B6 and B7 share an identical storage budget; B1, B2 and B4 do not, and their cost columns show what their margin is bought with. 95% intervals and all remaining conditions are in the appendix.

failure condition	system	KS	accuracy	ISS	contr. F1	storage/claim	msgs/claim
50% random churn	B1	0.730	0.499	0.833	0.781	36.328	42.475
50% random churn	B2	0.876	0.605	1.000	0.900	193.749	209.462
50% random churn	B3	0.820	0.562	0.622	0.459	4.000	8.000
50% random churn	B4	0.864	0.586	0.960	0.896	48.437	99.588
50% random churn	B5	0.820	0.551	0.530	0.000	4.000	8.000
50% random churn	B6	0.821	0.574	0.686	0.638	4.000	8.000
50% random churn	B7	0.881	0.763	0.686	0.486	4.000	14.517
50% loss, whole organisations	B1	0.789	0.539	0.900	0.843	36.328	41.998

failure condition	system	KS	accuracy	ISS	contr. F1	storage/claim	msgs/claim
50% loss, whole organisations	B2	0.877	0.605	1.000	0.899	193.749	209.472
50% loss, whole organisations	B3	0.820	0.561	0.622	0.462	4.000	8.000
50% loss, whole organisations	B4	0.787	0.494	0.750	0.772	48.437	102.565
50% loss, whole organisations	B5	0.820	0.550	0.530	0.000	4.000	8.000
50% loss, whole organisations	B6	0.822	0.575	0.687	0.639	4.000	8.000
50% loss, whole organisations	B7	0.883	0.764	0.687	0.489	4.000	14.517
50% targeted lineage attack	B1	0.876	0.600	1.000	0.937	36.328	41.286
50% targeted lineage attack	B2	0.870	0.597	0.993	0.895	193.749	210.089
50% targeted lineage attack	B3	0.678	0.441	0.504	0.351	4.000	8.000
50% targeted lineage attack	B4	0.749	0.472	0.691	0.747	48.437	105.090
50% targeted lineage attack	B5	0.677	0.429	0.437	0.000	4.000	8.000
50% targeted lineage attack	B6	0.679	0.451	0.550	0.506	4.000	8.000
50% targeted lineage attack	B7	0.726	0.594	0.550	0.398	4.000	14.517
20% origin-source corruption	B1	0.675	0.431	0.933	0.870	36.328	41.763
20% origin-source corruption	B2	0.719	0.464	1.000	0.894	193.749	205.168
20% origin-source corruption	B3	0.702	0.456	0.706	0.628	4.000	8.000
20% origin-source corruption	B4	0.719	0.461	0.994	0.901	48.437	96.845
20% origin-source corruption	B5	0.695	0.444	0.561	0.000	4.000	8.000
20% origin-source corruption	B6	0.717	0.475	0.810	0.827	4.000	8.000
20% origin-source corruption	B7	0.770	0.640	0.810	0.794	4.000	14.517

Figure captions

Figure 1. Knowledge survival and task accuracy as failure escalates (N = 10,000, 30 seeds, bands are 95% CIs). Columns are three failure structures at matched node-level severity: uniform random churn, correlated loss of whole organisations, and a white-box adversary that knows each system's placement and removes the hosts of its scarcest surviving independent evidence first. Axes span the full [0, 1] range. The equal-budget policies (B3, B5, B6) separate on accuracy rather than on availability; the systems that stay flat at extreme severity (B2, B4) do so at one to two orders of magnitude more storage, quantified in Figure 2.

Figure 2. Resilience–efficiency frontier under the 50% white-box attack (N = 10,000). Points are the eight systems; dashed lines sweep the storage budget $k \in \{1, 2, 4, 8, 16, 32\}$ for the three equal-budget policies, so the comparison does not depend on one arbitrary budget. Left: knowledge survival against storage cost. Right: against storage plus communication. Error bars are 95% CIs; the x-axis is logarithmic because the systems differ by two orders of magnitude in cost.

Appendix table

Table A1. Storage-budget sweep under the 50% white-box attack (N = 10,000, 15 seeds). Each equal-budget policy is run at $k \in \{1, 2, 4, 8, 16, 32\}$ published replicas per claim. Knowledge survival converges as k grows — availability is set by replica count — while independent-support survival does not: the diversity-blind policy saturates well below the lineage policy at every budget, because additional replicas of one origin add no independent support.

system	budget k	storage/claim	msgs/claim	KS	accuracy	ISS
B3	1	1.000	2.000	0.205	0.115	0.133
B3	2	2.000	4.000	0.435	0.261	0.297
B3	4	4.000	8.000	0.679	0.442	0.504
B3	8	8.000	16.000	0.829	0.567	0.689
B3	16	16.000	30.936	0.875	0.606	0.813
B3	32	32.000	49.350	0.877	0.606	0.892
B5	1	1.000	2.000	0.207	0.113	0.133
B5	2	2.000	4.000	0.438	0.256	0.283
B5	4	4.000	8.000	0.677	0.430	0.437
B5	8	8.000	16.000	0.826	0.557	0.533
B5	16	16.000	30.976	0.872	0.598	0.563
B5	32	32.000	49.691	0.876	0.602	0.566
B6	1	1.000	2.000	0.205	0.119	0.133

system	**budget k**	**storage/claim**	**msgs/claim**	**KS**	**accuracy**	**ISS**
B6	2	2.000	4.000	0.438	0.270	0.315
B6	4	4.000	8.000	0.679	0.452	0.550
B6	8	8.000	16.000	0.829	0.578	0.773
B6	16	16.000	30.961	0.873	0.616	0.918
B6	32	32.000	49.618	0.876	0.618	0.982

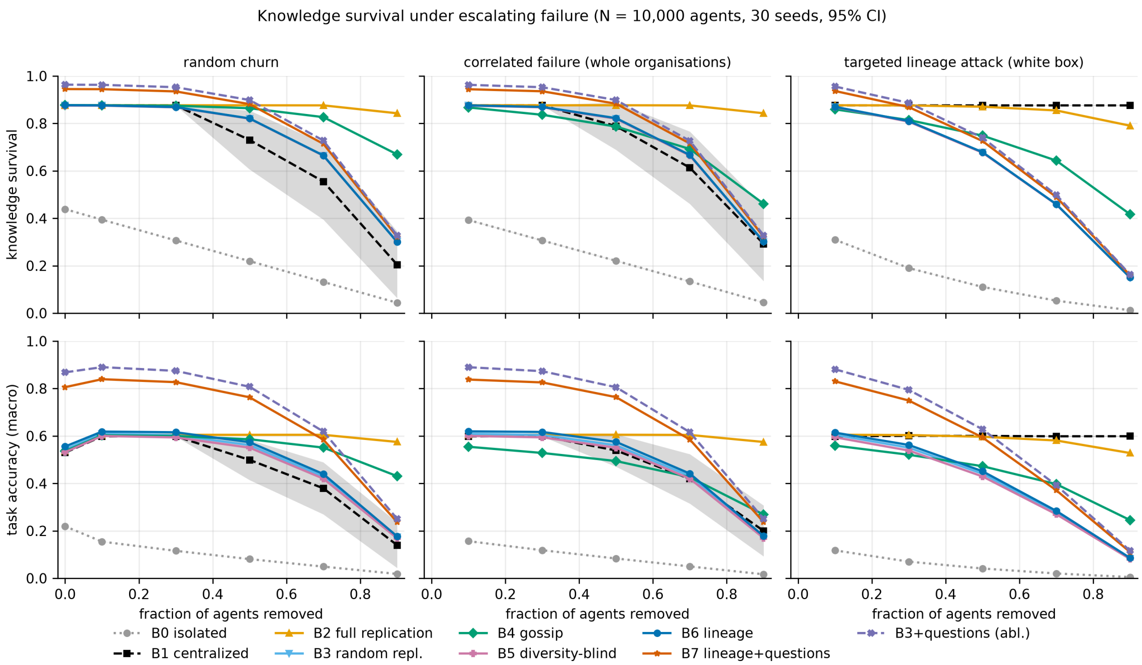

Figure 1: knowledge survival and task accuracy as failure escalates.

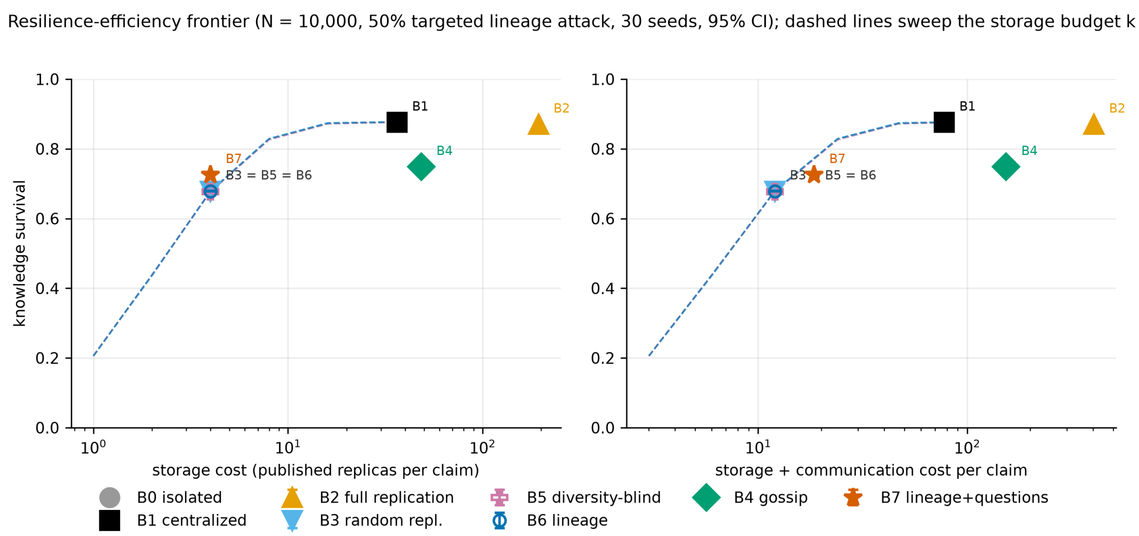

Figure 2: resilience-efficiency frontier under the white-box attack.